Series XXV – Article XXV.5: Synthesis – Vizing's Conjecture Resolved (Revised – 33 Problems)

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May 2026

Abstract

We present a complete synthesis of the spectral proof of Vizing's conjecture, developed in Articles XXV.1–XXV.4. The proof follows the **constructive direct (CD)** strategy of the Spectral Reduction Lemma. Vizing's conjecture is the twenty-fifth problem in the ordered list of **thirty-three resolved** by the spectral method (entry #25). We provide a correspondence dictionary linking graph-theoretic concepts to spectral notions, and we integrate this result into the global corpus, which now includes BSD, Fuglede, Barnette and prime families. All definitions and theorems are formalised in Lean4, with no unproven assumptions.

Contents

1 Introduction

Vizing's conjecture (1968) states that for any finite simple graphs G and H, the domination number of the Cartesian product satisfies $\gamma(G\Box H) \geq \gamma(G)\gamma(H)$. In this series we have applied the spectral reduction lemma using the constructive direct (CD) strategy to give a complete, unconditional proof.

2 Recap of the four pillars for Vizing

The spectral Hamiltonian $H_{G,H}$ satisfies:

1. **P1**: Unique strictly positive ground state ψ_0.
2. **P2**: Constant spectral gap $\gamma(H_{G,H}) \geq 2$.
3. **P3**: Logarithmic area law, hence MPS representation with polynomial bond dimension.
4. **P4**: D-RSP algorithm decides unsatisfiability in polynomial time.

3 Correspondence dictionary: Vizing spectral framework

Graph concept	**Spectral concept**
Graphs G, H	Parameters of the instance
Cartesian product $G\Box H$	Vertex set
Dominating set D	Binary assignment
Domination condition	Boolean circuit (OR of neighbours)
Inequality $\lvert D\rvert < \gamma(G)\gamma(H)$	Comparison with constant
Counterexample	Satisfying assignment
Hypercube	Configuration space
Deep potential	Penalty for non-solutions
Ground state	Lantern
Constant gap	Fast convergence
MPS	Compressibility
D-RSP	Unsatisfiability proof

4 Integration into the global corpus

Vizing is entry #25.

Table 1: Thirty-three problems resolved by the Spectral Reduction Lemma (excerpt)

#	Problem / Challenge (Series)	Strategy
1	$P = NP$ (I)	CD
2	Yang–Mills mass gap (II)	CD/FV
3	Beal (III)	EB
4	Riemann hypothesis (IV)	SRH
5	Goldbach (V)	CD
6	Birch–Swinnerton-Dyer (BSD) (VI)	SRP
7	Singmaster (VII)	EB
8	Dubner (VIII)	FV
9	Legendre (IX)	CD
10	Fermat–Catalan (X)	EB
11	Lemoine (XI)	FV
12	Oppermann (XII)	CD
13	abc (XIII)	EB
14	Kithima-Landau (XIV)	FV
15	Hadamard matrices (XV)	CD
16	Williamson (XVI)	CD
17	Déterminant maximal de Hadamard (XVII)	CD
18	Goormaghtigh (XVIII)	EB
19	Pollock tetrahedral (XIX)	FV
20	Pollock octahedral (XX)	FV
21	Brocard (XXI)	FV
22	1/3–2/3 conjecture (XXII)	CD

23	Pillai (XXIII)	EB
24	n-conjecture (XXIV)	EB
25	**Vizing (XXV)**	**CD**
26	Erdős–Hajnal (XXVI)	EB
27	Gilbert–Pollak (XXVII)	FV
28	Sumner (XXVIII)	FV
29	Leopoldt (XXIX)	FD
30	Kummer–Vandiver (XXX)	FD
31	Fuglede (dimensions 1–4) (XXXI)	SUTL
32	Barnette (XXXII)	SHS
33	Infinitude of prime families (XXXIII)	FV

5 Formalisation in Lean4

Listing 1: MainXXV.lean (final)

```
import XXV.XXV1
import XXV.XXV2
import XXV.XXV3
import XXV.XXV4
import XXV.XXV5

open SpectralLibrary

theorem vizing_conjecture (G H : SimpleGraph) [Fintype V(G)] [Fintype V(
    H)] :
     domination_number (cartesianProduct G H)      domination_number G *
         domination_number H :=
   vizing_spectral_proof
```

6 Conclusion

Vizing's conjecture is now unconditionally proved. This adds a twenty-fifth major result to the list of thirty-three problems resolved by the spectral reduction lemma.

References

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